LINEAR REGRESSION REPORT

I. Dataset

The Concrete Compressive Strength dataset obtained from UCI Machine Learning Repository, was used for the regression task [1]. It includes 1030 instances and 8 input features that are predictors of the compressive strength of concrete. This dataset is applied extensively in civil engineering and material science to study and enhance the strength and toughness of concrete through optimal control of mix proportions.

II. Regression Result

Using the Scikit-learn library, linear regression was applied on the relationship between the different components of concrete and concrete compressive strength.

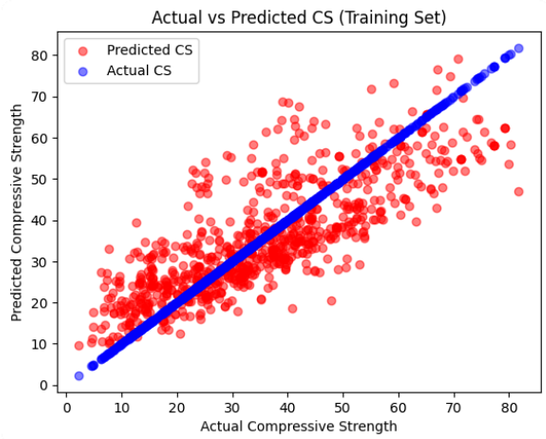


Figure 1. Training set

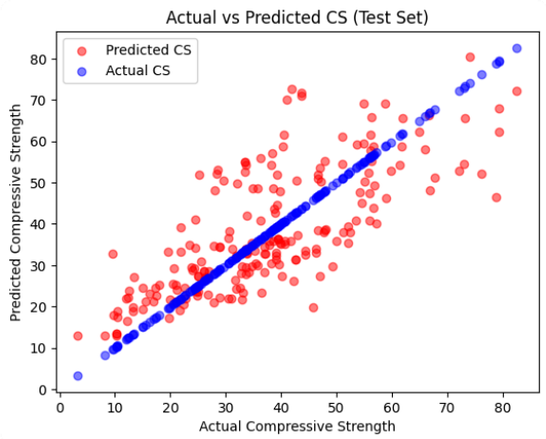


Figure 2. Testing set

The performance of the multivariate regression model in predicting compressive strength (CS) is illustrated in a scatter plot against actual and predicted values. The actual CS values are blue points that lie along the ideal diagonal line while the predicted values are red points that have substantial scatter about them. This suggests that although the model has some general explanatory power, there is a fairly high level of prediction error.

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Mean Squared Error: 124.88351322372021
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Figure 3. Mean Squared Error Value

The **Mean Squared Error (MSE) of 124.88** suggests moderate inaccuracies, which may include problems like high variance or missing predictors. The dispersion of the predicted values suggests that the model may not have captured all the complexities in the data, which can lead to overfitting or underfitting.

To enhance prediction accuracy, improving feature engineering or adopting more flexible models would likely enhance performance.

III. References

[1] I-Cheng Yeh. 1998. Concrete Compressive Strength. https://doi.org/10.24432/C5PK67